Vijay Baskar

Head of Software Technologies @ Retailetics | AI/ML Developer | Mobile Systems & Scaled Engineering

Specializations: Android Architecture | Kotlin | Cloud-Connected Apps | AI/ML

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SUMMARY

Senior Android Engineer with 12+ years of experience building high-performance mobile apps for millions of users. Expertise in Kotlin, Jetpack Compose, and Android architecture, with a proven track record of reducing ANR crashes by 70% and leading technical initiatives. Full Stack Engineer with additional expertise in scalable solutions:

Android Development:
- Delivered 20+ production apps using Android SDK, Jetpack components
- Reduced ANR crashes by 70% through architecture optimization
- Improved app performance by 30% via profiling and tuning

Cross-Platform & Backend:
- Built apps with Flutter and Kotlin Multiplatform
- Designed microservices (Spring Boot) handling 500K+ users
- Optimized backend response time by 40%

AI/ML Integration:
- Developed AI-powered self-checkout saving $2M+
- Implemented on-device ML with TensorFlow Lite
- Created indoor navigation replacing Bluetooth beacons

Engineering Leadership:
- Led design reviews and mentored junior engineers
- Established CI/CD pipelines (Jenkins, Fastlane)
- Implemented automated testing (JUnit, Espresso)

Seeking to leverage my expertise in Android architecture, performance optimization, and AI integration to build transformative mobile experiences.

TECHNICAL SKILLS

Programming Languages: Java, Kotlin, Dart, Python, MATLAB
Android Development: Android SDK, Jetpack Compose
Cross-Platform Development: Flutter, Dart, Kotlin Multiplatform (KMP), iOS
Design Principles & State Management: SOLID, Clean Architecture, MVVM, MVP, BLoC, Provider
Deep Learning Frameworks: TensorFlow, PyTorch, Keras
Libraries & Tools: NumPy, Pandas, Scikit-learn, OpenCV, NLTK, Git, Docker
Neural Networks: DNN, CNN, RNN, LSTMs, Transformers, GANs
Natural Language Processing (NLP): Text Pre-Processing, Sentiment Analysis, Language Modeling
Backend Development & APIs: RESTful APIs, GraphQL, WebSockets, Spring Boot, Postgres, SQLite
DevOps & Tools: Jenkins, Fastlane, Git, GitHub, Bitbucket, GitLab, JUnit, Espresso, Flutter Test
IoT & Hardware: Arduino, Raspberry Pi, Orange Pi, Bluetooth Beacons, IMU Sensors
In-store / Retail Systems: ezyCart, Self-checkout kiosk, AI recipe builder, NFC / Tap to Pay, Product finder, Voice assist, Stock audit, ezyList, Flutter SDK lite

SYSTEM DESIGN & ARCHITECTURE

High-Performance Mobile Architecture: Optimized Android app performance, reducing ANR crashes by 70%
Scalable Backend Systems: Designed microservices handling 500K+ users, reducing response time by 40%
Database Optimization: Improved query performance by 60% with indexing, caching, and pagination
Socket Communication & WebSockets: Implemented real-time data streaming in Android using Kotlin & WebSockets

AI & MACHINE LEARNING EXPERTISE

Computer Vision: Built real-time object detection models with OpenCV & TensorFlow
NLP: Implemented sentiment analysis & chatbot AI using Transformer models
AI for Mobile: Integrated on-device ML models for predictive text & recommendation engines
Optimized AI Models: Reduced inference time by 30% using model quantization techniques

EXPERIENCE & PROJECTS

Retailetics SDN BHD - Head of Software Technologies **February 2021 – Present**

Malaysia, KL

- Retailetics AI-infused Self-Checkout Smart Trolley (ezyCart). I spearheaded the development of an AI-powered self-checkout shopping cart, streamlining the checkout process for a seamless, queue-free experience. This involved architecting the system, leading technical discussions with stakeholders, and designing the user experience. Additionally, I developed a real-time data transfer pipeline using sockets between IoT devices and the ezyCart Android app, ensuring fast data synchronization while eliminating slow transfers and redundant permission requests.
- Shipped the self-checkout kiosk on the same stack — scan, bag on the trays, and pay without a cashier line. Integrated multiple payment gateways, including Tap to Pay and NFC, so checkout works on cart and kiosk.
- Built an AI recipe system on ezyCart: as items go in, the cart suggests recipes by cuisine, plays the cooking video, and lets shoppers add missing products to the basket. In-aisle upselling and cross-selling through ads in the same cart system.
- Delivered Million-Dollar Savings with AI-powered Indoor Navigation System. Saved over $2M by developing an AI-powered indoor navigation system, eliminating the need for thousands of Bluetooth beacon devices, resulting in significant cost savings and improved efficiency. Product finder: search a SKU and navigate to its location in the aisle.
- Stock auditing on the smart cart so the floor can count, scan, and flag gaps without a separate handheld round. Voice Assist AI for customer queries, product facts, and promo info. Monitoring and troubleshooting for every active cart and self-checkout station.
- Conceptualized, Designed, and Developed Cross-Platform Support Staff App (Flutter) & Android TV Cart Monitor System. Flutter SDK lite of ezyCart so small vendors can embed scan-and-pay in their own phone and tablet apps.
- ezyList: shared shopping lists with friends and family, nearby stores with price and availability, navigate to the store, purchase in one scan, digital receipt, and member-only promos.
- Conceptualized and Developed “Lisa” – An AI Shopping Assistant that resolves customer queries and provides real-time product recommendations and promotions, enhancing the in-store shopping experience.
- Developed and Deployed Machine Learning Models for Various Industries.
- Automated Build Pipeline and Deployment for Various Merchants and Outlets. Led the automation of build pipelines and deployment processes, streamlining operations and reducing deployment time across multiple merchants and retail outlets.
- Led Split Testing Initiatives Driving Multi-Million Dollar Revenue Growth. Spearheaded split testing initiatives that generated over $X million in additional revenue for Retailetics, significantly enhancing customer experiences and exceeding annual performance goals.

Colan infotech PVT LTD - Senior Software Engineer **Nov 2012 – Feb 2021**

Chennai, India

- Delivered 20+ High-Performance Mobile Apps (Android/iOS). Developed native (Kotlin/Java) and cross-platform (Flutter) applications, showcasing expertise in Android (Java, Kotlin, SDK, MVVM, Jetpack Compose) and Flutter (State management, widgets, animations, integrations). Implemented automated CI/CD pipelines (Jenkins, GitLab, Bitrise) for seamless deployments and architected scalable mobile backends using AWS, Firebase, and RESTful APIs.

TAN Intelligence - Founder

Independent venture · AI-powered forex trading indicator

- TAN Intelligence is an independent company building an original Artificial Intelligence trading indicator for the forex market. Direction: indicator, then signals, then an automated bot with explicit risk controls. Founder work, not a second employer.

EDUCATION

Anna University / GGR College of Engineering Vellore, India

Bachelor Of Engineering in Computer Science *Jun 2007 – Nov 2011*

Target Software Solution Chennai, India

Mobile Application Development using Android and iOS *Jan 2012 – Oct 2012*

SOLID Principles — Coursera · AI/ML Architecture — IBM · Master prompting — Claude and Coursera